

Analysis of Saffman-Taylor experiments in shear-thinning fluids

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1 Summary

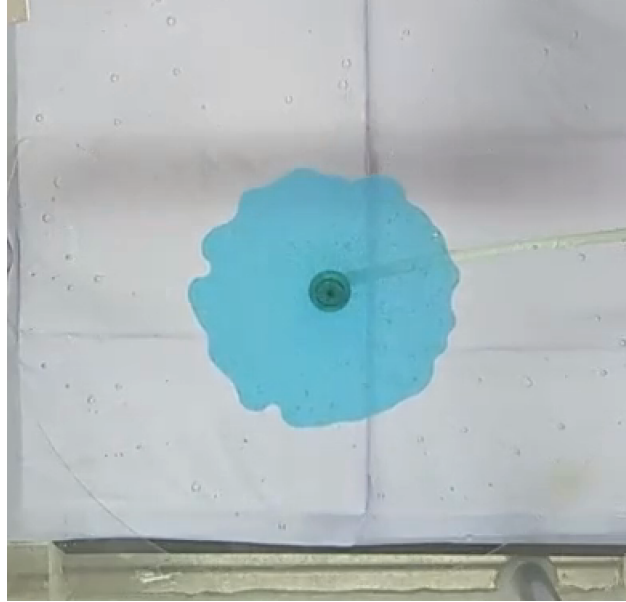


Figure 1: Xanthan gum (blue) injected into silicone oil (clear), showing slight Saffman-Taylor fingering.

One of the possible improvements to the mathematical model derived in Hutchinson (2025) is the addition of wall slip of the shear-thinning Xanthan gum owing to the presence of a thin boundary layer of Newtonian silicone oil. To determine whether this was the case, the recordings of the experiments were analysed to compare the theoretical volume of the Xanthan gum in the case of no boundary layers with the actual volume injected into the cell. If there is only a slight discrepancy, the boundary layers are small and we may simply alter the boundary conditions on the walls of the Hele-Shaw cell. However, if the volume discrepancy is large, then the boundary layers themselves may need to be modelled.

1.1 Key findings

1. The measured volumes were greater than that injected in all but one experiment, implying the existence of silicone oil boundary layers.
2. The greatest discrepancy was observed in the experiment with the most pronounced instabilities (and therefore surface area to volume ratio). This is consistent with the hypothesis of boundary layer formation, as a higher surface area to volume ratio caused by the instability would allow more silicone oil to intrude into the Xantham gum.
3. In this most extreme case, the boundary layers would have a uniform thickness of approximately 0.05 mm, or 10 % of the half-width of the Hele-Shaw cell.
4. However, this method of volume estimation is highly sensitive to the centimetre-to-pixel conversion factor, α , used, scaling like α^2 . Therefore, future experiments might aim to better constrain this value by placing multiple objects of known lengths in the video frame to check internal consistency of the α value used in each experiment.

2 Methods

2.1 Pre-processing and edge detection

First, the experiment video is loaded, cropped and the red colour channel is selected (the blue gum and white background have the greatest contrast in the red channel). Then, the centimetre-to-pixel conversion factor, α , is calculated by measuring the number of pixels across the diameter of the outer injection nozzle, whose diameter was measured to be 1.8 cm in the lab (Figure 2).

With the video loaded and calibrated, the edge of the gum is detected at each frame in the video. To do this, we first blur the image (using Gaussian blur). Then, we pass the blurred image through a Sobel filter, which approximates the gradient of the image lightness. This filter emphasises the edges in the image; however, it does not produce clear, crisp edges and creates many edge artefacts in the image (since it simply applies a gradient operator to the entire image). The image is then passed through a watershed segmentation algorithm to more unambiguously segment the gum from the background and other artefacts. To do this, we select lightness values that are unambiguously points in the gum and in the background. This algorithm treats the output of the Sobel filter as though it were a geological map, and aims to find the lines running along the tops of the ‘ridges’ corresponding to the emphasised edges between the gum and the background. The output of this algorithm is then filled and all objects smaller than some critical radius are removed from the image, leaving only the Xantham gum. We then apply the Canny edge detection algorithm to this image to produce a set of clear, crisp edges around the gum (Figure 3)

2.2 Volume and Perimeter calculation

To calculate the volume of the gum, we fill in the edges of the gum and multiply the number of True pixels by α^2 . Calculating the perimeter of the gum is more involved: if one simply

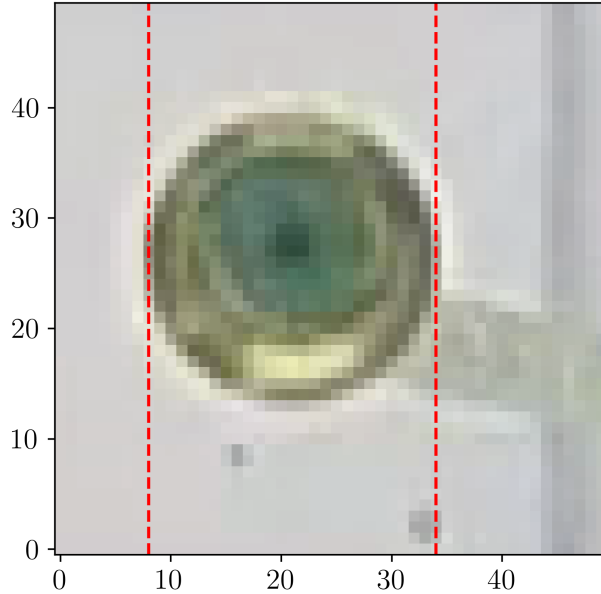


Figure 2: The value of α was determined using the known diameter of the outer injection nozzle in each experiment.

sums the distances between edge points, then the perimeter will be overestimated by up to a factor of $\sqrt{2}$ for edges oriented at 45° (e.g. the staircase-like points in the lower right of Figure 3). To overcome this, we take the Fourier transform of the edge points and perform a low-pass filter, filtering out the high-wavenumber modes associated with the noise at the gum edge before transforming back.

This processing step produces a large improvement in the perimeter estimation of the curve (Figure 5). However, even without de-noising the curve, the peak increase in curve complexity (measured by the ratio of the measured perimeter of the gum to the perimeter of a cylinder of the same volume) occurs at the same time, though the peak is less pronounced. Integrating the denoised curve produces practically no effect on the volume estimation compared to simply multiplying the number of gum pixels by α^2 . This is expected given the definition of an integral as the limit of a sum; however, it is a nice sanity check (Figure 6).

3 Results

The experiments are labelled according to their appearance in the results tables in Hutchinson (2025).

3.1 Table 1, Experiment 1

Figure 6 shows the volume results for this experiment, and Figure 7. Note that the perimeter ratios are plotted against nondimensional time, using the following scaling from Hutchinson

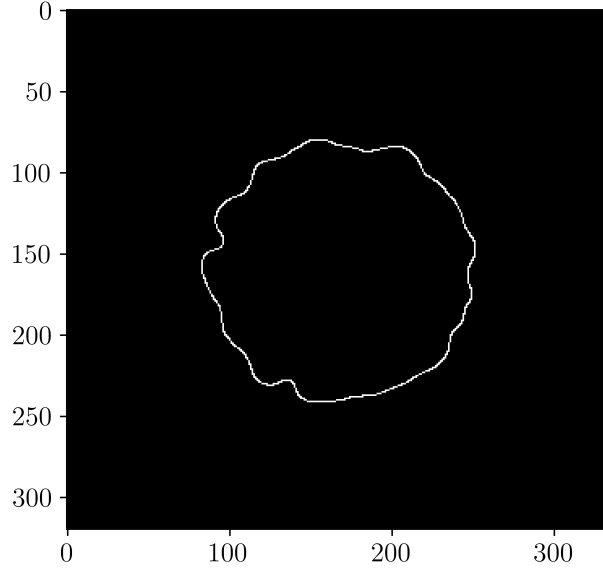


Figure 3: The output of the edge detection workflow. The Xantham gum edge is clear, crisp and continuous, with no gaps or sharp discontinuities. This produces a Boolean matrix, with the edges indicated by True values.

Experimental Parameter	Value
a_0	0.5 cm
$2H$	0.1 cm
Q	$2 \text{ cm}^3 \text{ s}^{-1}$
μ_N	1 Pa s

Table 1: Experimental parameters for table 1, experiment 1.

(2025):

$$T = \frac{t}{4\pi H a_0^2 / Q},$$

where the values of the experimental parameters are those in Table 1.

The measured gum volume closely tracked the injected volume, although it decreased slightly over time. However, this may be due to effects such as poor edge detection (artefacts). Moreover, varying the α value (labelled cmppx in the figure) by just 5% shifts the measured volume above or below the injected volume line. The lower panel shows the thickness of the silicone boundary layer, with a maximum boundary layer thickness of approximately 0.05 mm at late times.

3.2 Table 1, Experiments 2, 3 and 4

3.2.1 Experiment 2

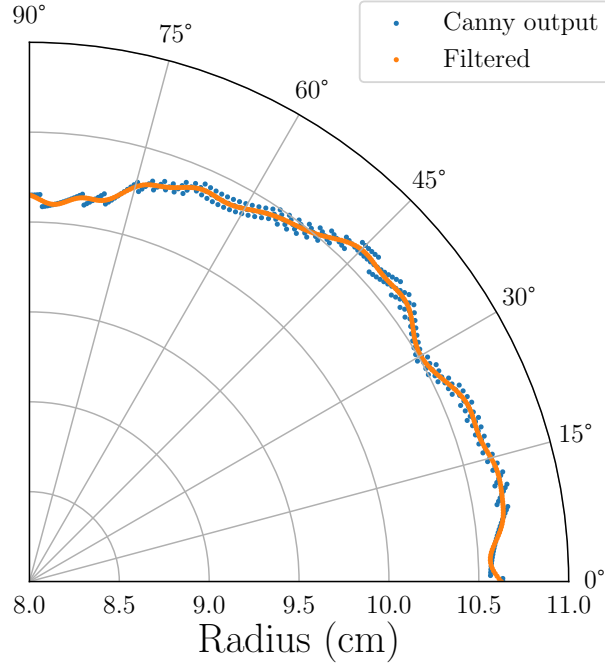


Figure 4: A zoomed section of an example curve, showing the smoothed (denoised) curve created using low-wavenumber modes.

Experimental Parameter	Value
a_0	0.8 cm
$2H$	0.1 cm
Q	$2 \text{ cm}^3 \text{ s}^{-1}$
μ_N	1 Pa s

Table 2: Experimental parameters for table 1, experiment 2.

3.2.2 Experiment 3

Experimental Parameter	Value
a_0	1 cm
$2H$	0.1 cm
Q	$2 \text{ cm}^3 \text{ s}^{-1}$
μ_N	1 Pa s

Table 3: Experimental parameters for table 1, experiment 3.

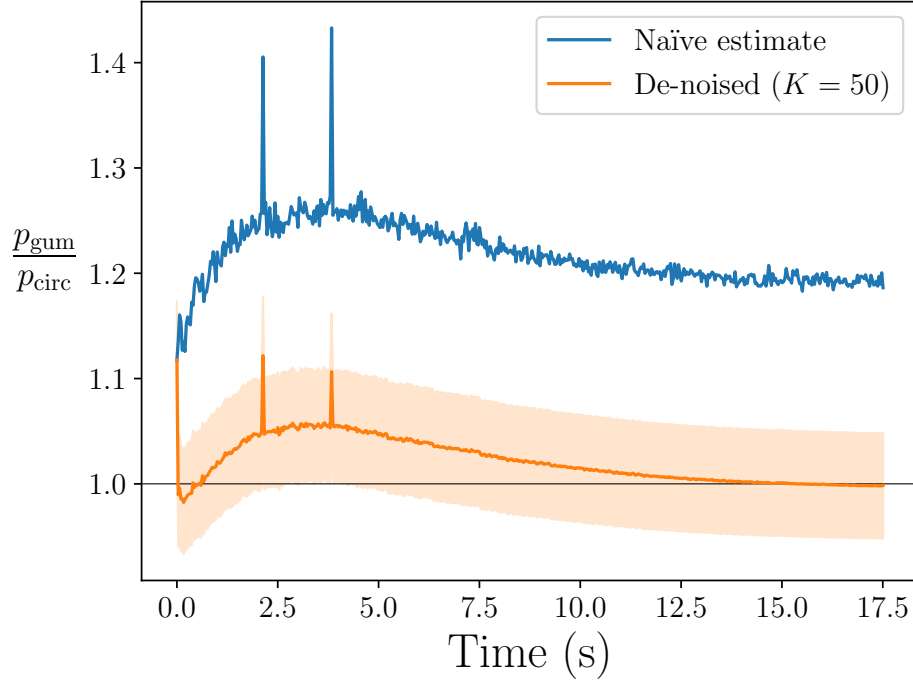


Figure 5: Ratio of the perimeter of the Xantham gum to the perimeter of a circle of the same volume (Table 1, Experiment 1). Denoising the curve produces a lower, more accurate perimeter than the naïve perimeter estimation method of summing distances between adjacent points. In particular, the denoised curve tends to 1, as one would expect as the gum tends towards axisymmetry.

3.2.3 Experiment 4

Experimental Parameter	Value
a_0	1.8 cm
$2H$	0.1 cm
Q	$2 \text{ cm}^3 \text{ s}^{-1}$
μ_N	1 Pa s

Table 4: Experimental parameters for table 1, experiment 4.

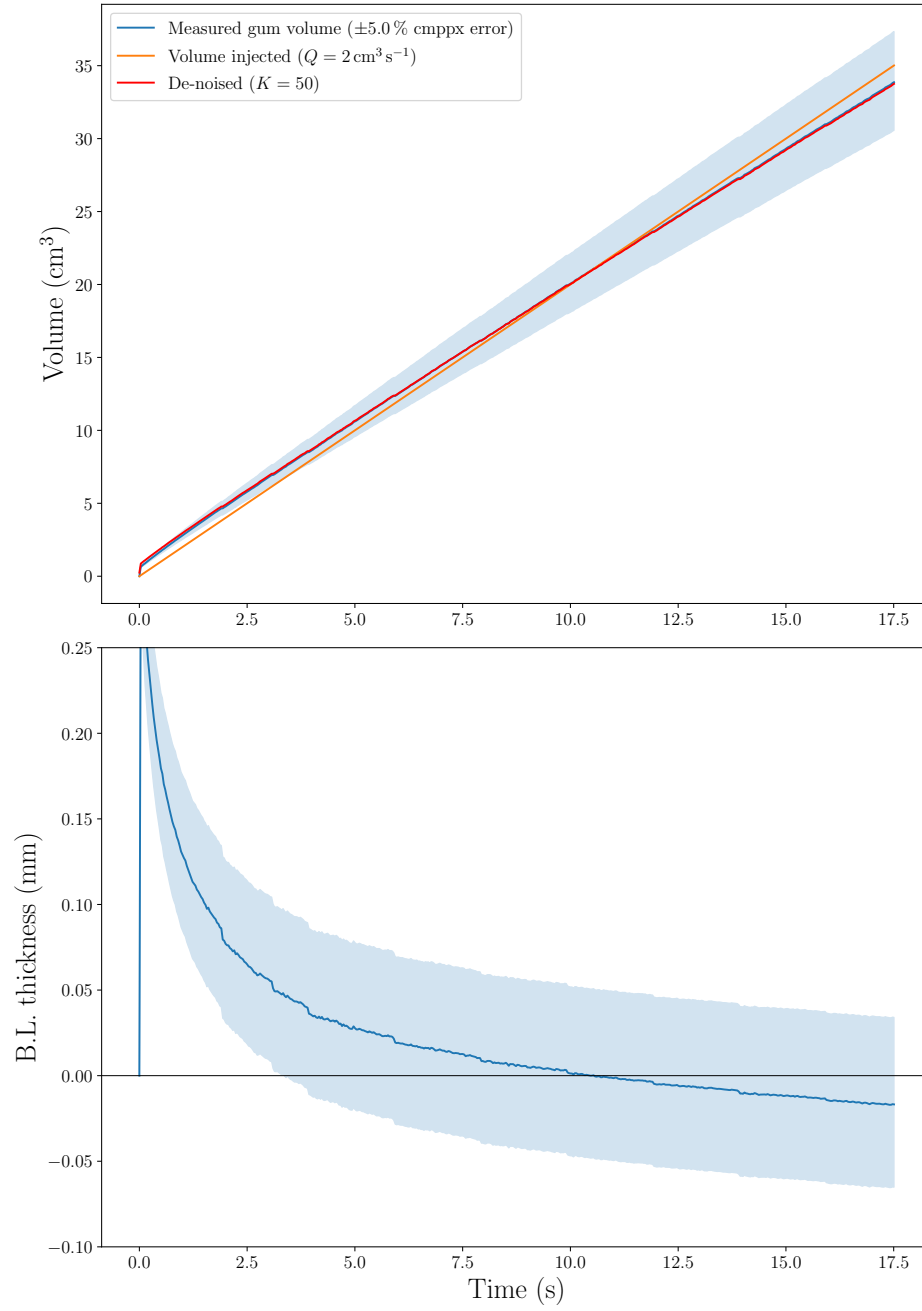


Figure 6: Volume and theoretical silicone boundary layer thickness (assuming uniform BL thickness across the entire gum) for Table 1, Experiment 1. Note that the volume estimation using the pixel counting method (blue) is almost identical to the method of integrating the denoised gum edge (red).

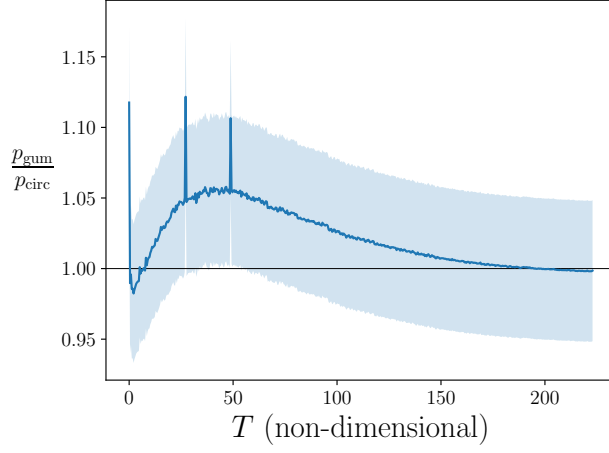


Figure 7: Table 1, Experiment 1: Ratio of gum edge perimeter to that of an axisymmetric gum layer of the same volume, plotted against nondimensional time.

3.3 Table 2, Experiment 1

Experimental Parameter	Value
a_0	1.6 cm
$2H$	0.1 cm
Q	$2 \text{ cm}^3 \text{ s}^{-1}$
μ_N	4.85 Pa s

Table 5: Experimental parameters for table 2, experiment 1.

This experiment showed the most pronounced instabilities, and unlike the other experiments, did not return to axisymmetric spreading at late times. Therefore, its perimeter plot does not tend to one as in the other experiments. This experiment showed the most pronounced difference in volumes, implying boundary layers of silicone oil of approximately 0.05 mm on the upper and lower surfaces of the Hele-Shaw cell (10 % of the cell half-width).

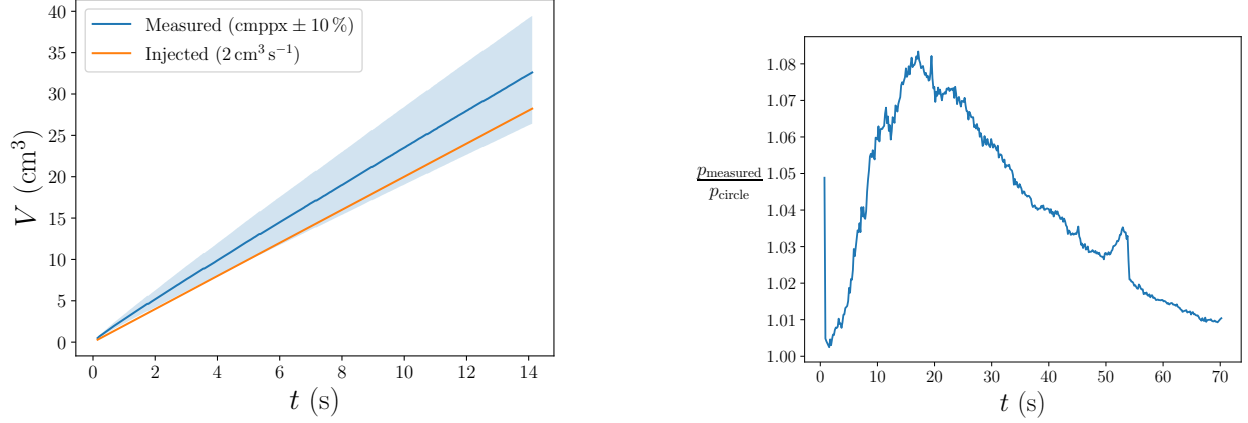


Figure 8: Table 1, experiment 2: (Left) Measured volume of Xanthan gum and volume injected. (Right) Ratio of measured perimeter of gum and the perimeter of a circle containing an equal volume of gum, plotted against nondimensional time. Note that $t = T = 0$ corresponds to the time of injection.

3.4 Table 2, Experiment 2

Experimental Parameter	Value
a_0	0.6 cm
$2H$	0.1 cm
Q	$2.6 \text{ cm}^3 \text{ s}^{-1}$ (plot suggests $2.4 \text{ cm}^3 \text{ s}^{-1}$)
μ_N	4.85 Pa s

Table 6: Experimental parameters for table 2, experiment 2.

This experiment showed almost perfect axisymmetric spreading, and the observed volume is in excellent agreement with the theoretical volume of an axisymmetric current injected at volumetric rate $Q = 2.4 \text{ cm}^3 \text{ s}^{-1}$ (Figure 15).

The pump appears to have stopped at late times. However, before this point, the gum shows close agreement with the injection volume, if slightly overestimated, implying some boundary layer formation.

4 Conclusions

These results show that the method of assuming the Hele-Shaw cell is entirely filled with Xanthan gum overestimates the gum volume in all but one experiment. This implies the existence of silicone oil boundary layers. This hypothesis is further supported by the greatest discrepancy being observed in the experiment with the most pronounced instability, likely owing to the significantly higher $p_{\text{gum}}/p_{\text{circ}}$ ratio in this case, allowing more silicone oil to intrude to form a boundary layer of approximately 0.05 mm. This estimate of a boundary layer taking up approximately 10% of the half-width of the Hele-Shaw cell may be taken as

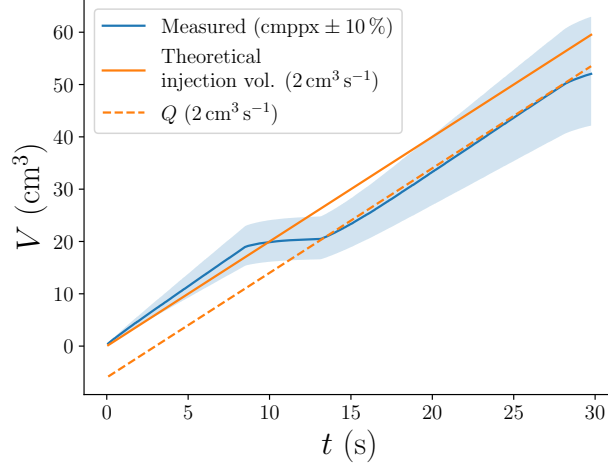


Figure 9: Table 1, Experiment 3 Volume plot.

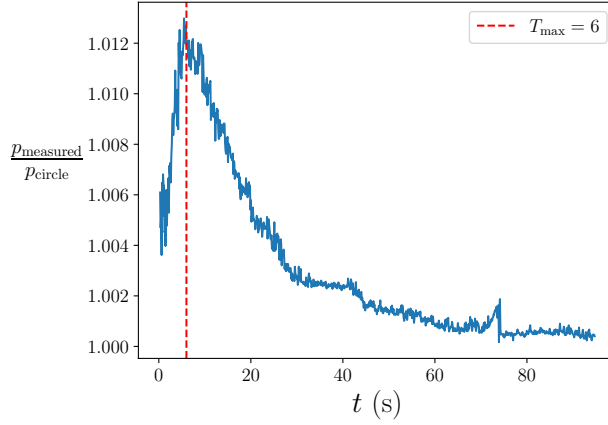


Figure 10: Table 1, Experiment 3 Perimeter plot.

the maximum likely thickness of the boundary layers in this experiment (note the boundary layers were assumed to be uniform thickness across the gum in this calculation).

Unfortunately, this method of inferring volumes is highly sensitive to the centimetre-to-pixel conversion factor, α , with the volume estimate varying like α^2 . Therefore, it is difficult to determine the boundary layer thickness with high accuracy. Future experiments may involve more checks (e.g. by placing multiple objects of known length in the frame to check the internal consistency of the α value chosen).

References

Hutchinson, A. (2025, November). Saffman-Taylor fingering instabilities in shear-thinning fluids displacing Newtonian fluids. *Applied Mathematical Modelling* 147, 116206.

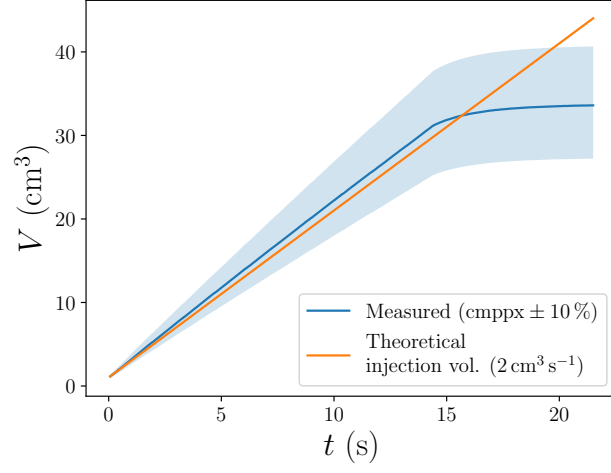


Figure 11: Table 1, Experiment 4 Volume plot

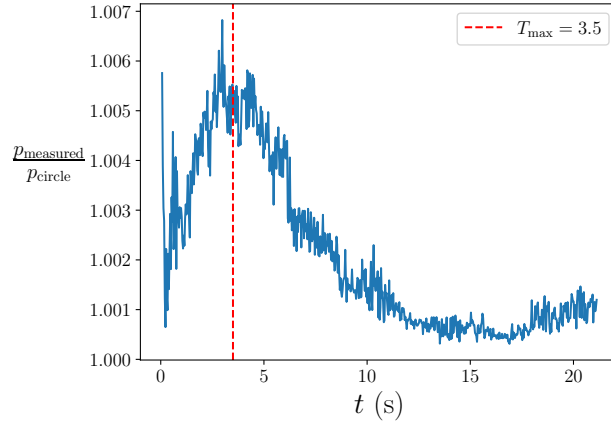


Figure 12: Table 1, Experiment 4 Perimeter plot

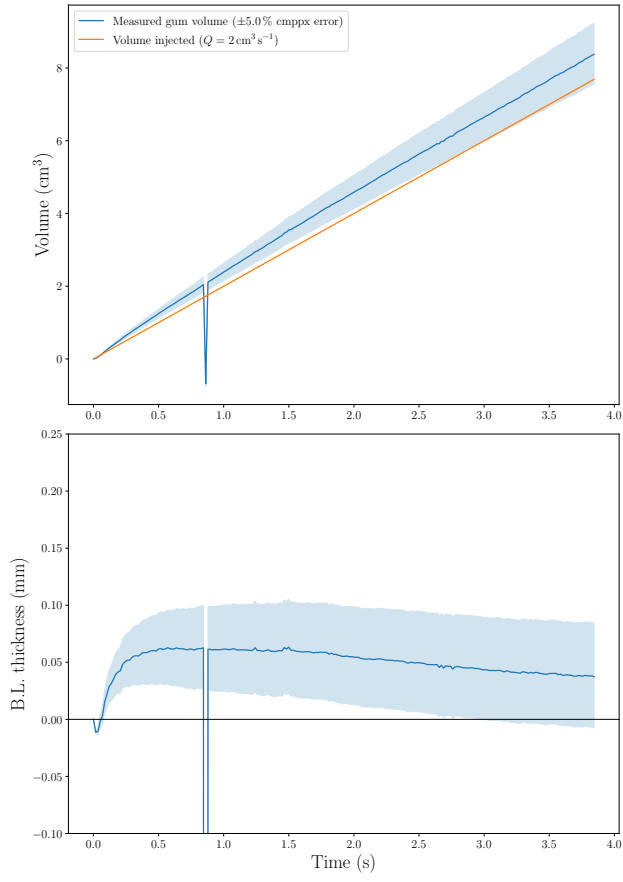


Figure 13: Table 2, Experiment 1 Volume plot.

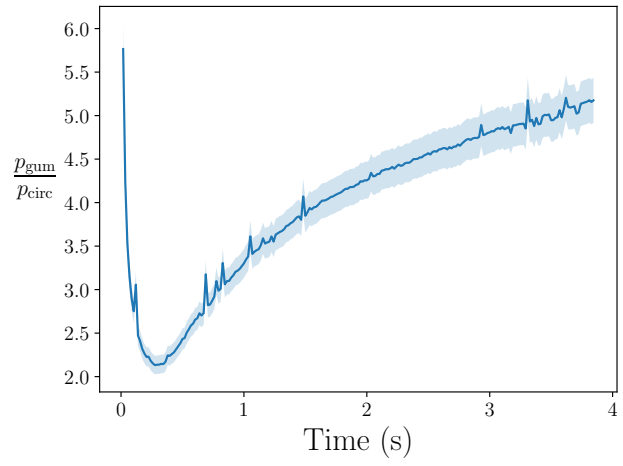


Figure 14: Table 2, Experiment 1, Perimeter plot.

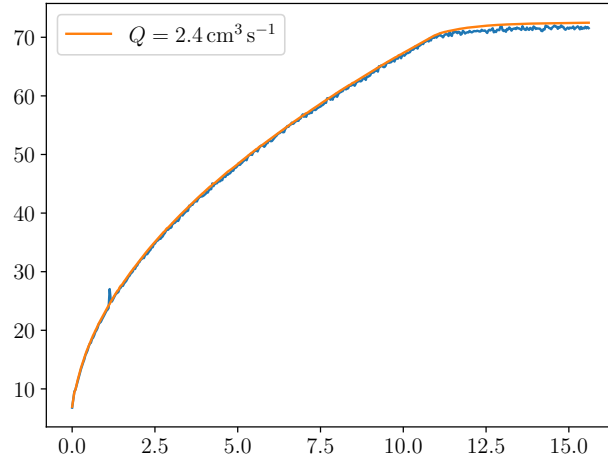


Figure 15: Measured gum perimeter and theoretical perimeter of axisymmetric spreading at injection rate $Q = 2.4 \text{ cm}^3 \text{ s}^{-1}$.

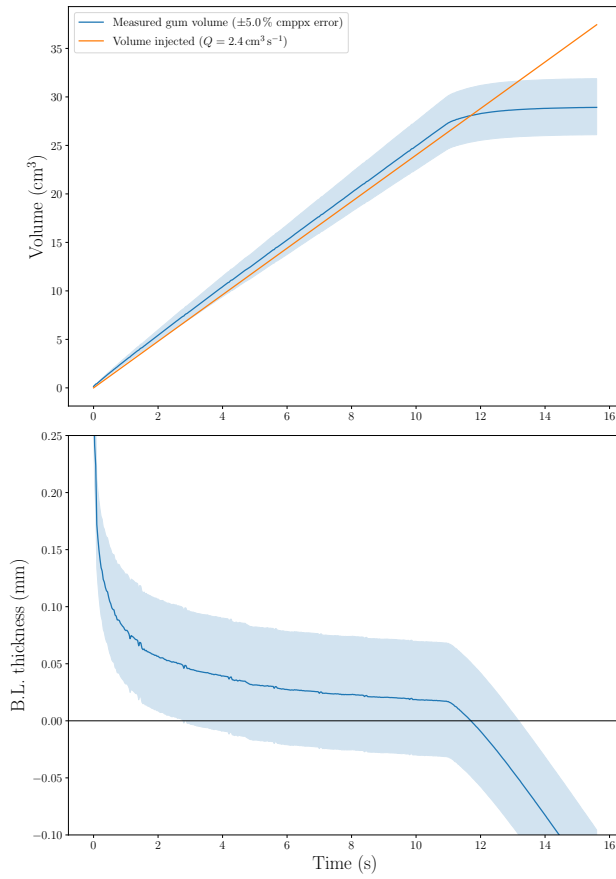


Figure 16: Table 2, Experiment 2 Volume plot.

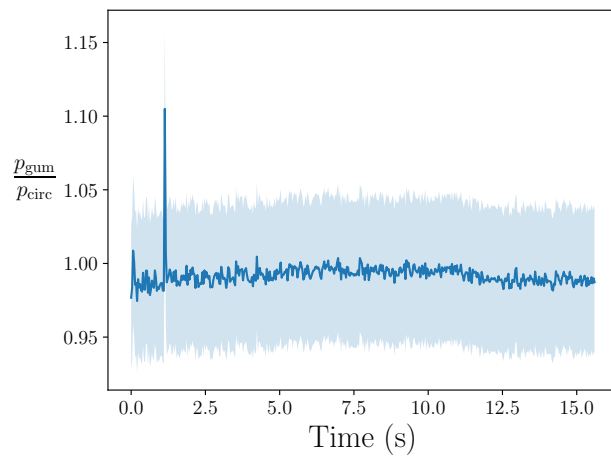


Figure 17: Table 2, Experiment 2 Perimeter plot.